Summary in Graph

Exam Summary (GO Classes GATE CS | Data Structures | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	10 (5 + 5)
Total Marks:	15 (5 + 10)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (32) out of 322 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Algorithms

If $g(n) = \Omega(n)$ and $h(n) = O(n)$ then $g(n).h(n)$ will be -

A. $\Theta(n)$

B. $O(n)$

C. $\Omega(n)$

D. None of these

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 02sec | Discuss

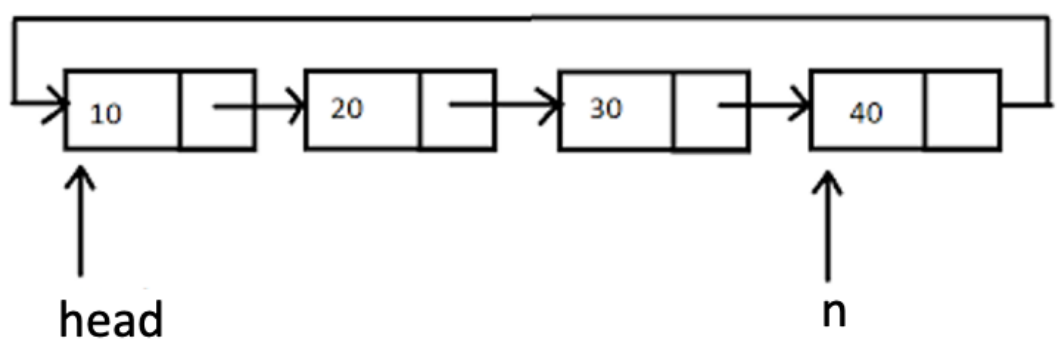
Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a linked list given in the figure below.

What is the value of n $\rightarrow$ next $\rightarrow$ next $\rightarrow$ next $\rightarrow$ data?

Where n and head are pointers to the following struct type and intitallised as per the figure shown.

```
struct node{
    int data;
    struct node *next;
};
```

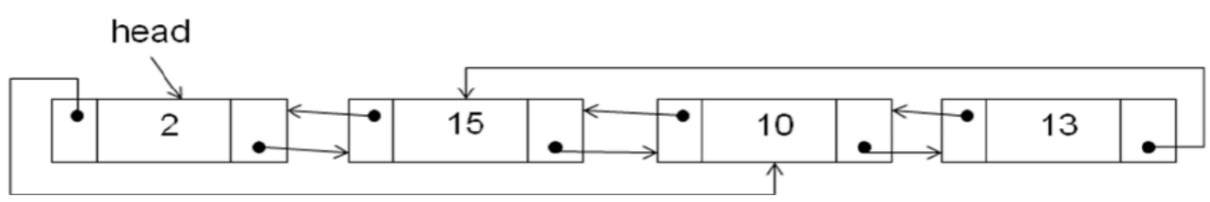


A. 10
B. 20
C. 30
D. 40

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming and DS

Consider the following linked list which has nodes of a doubly linked list type.



If we execute the following lines of code on a given linked list then what will be the output?

```
head->prev->next->next = head->next->next;
head = head->prev->prev;
head->prev = head;
printf("%d", head->data);
```

A. 2
B. 15
C. 10
D. 13

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a function ReverseList() which has following protype -

```
Node *ReverseList(Node *p)
```

Function reverse the nodes in a given linked list.

That is, head = ReverseList(head); will take a list held by head, reverse it and put back to head.

```
Node * ReverseList(Node * p) {
    if (!p) return NULL;
    if (p -> next) {
        Node * q = ReverseList(p -> next);
        p -> next -> next = p;
        p -> next = 0;
        return ______; //Line X
    }
    return ______; //Line Y
}
```

Fill in the blanks for Line X and Line Y. If we say Line X is filled by p then we consider Line X as **return p;** after filling the blank.
Line X and Line Y should be filled by which values respectively such that the function correctly reverses the given linked list.

A. p and $p \rightarrow$ next
B. p and q
C. $q \rightarrow$ next and p
D. q and p

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

The intent of the function below is to delete the last node of the list.

```
void removeLast(Node* first) {
    Node *p, *q;
    p = first;
    q = p->next;
    while (q->next != NULL) {
        p = q;
        q = q->next;
    }
    p->next = NULL;
    free(q);
}
```

Which of the following describes the class of all linked lists for which this function works correctly?

A. No linked lists
B. All non-empty linked lists
C. All linked lists with more than one node
D. The empty list and all linked lists with more than one node

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statement is/are FALSE? Assume that $\log$ base is 2 in all options.

A. $8^{\log(n^2)} \in \Theta\left(n^6\right)$

B. If $f, g,$ and h are functions from the natural numbers to the non-negative real numbers with $f(n) \geq g(n) \forall n \geq 1$, $f(n) \in \Theta(h(n))$, and $g(n) \in \Theta(h(n))$, then $(f(n) - g(n)) \in \Theta(h(n))$

C. $(\log n)^a$ is $\mathrm{O}\left(n^b\right)$ for any constant a, and for any constant $b > 0$

D. $T(n) = O(f(n))$ and $g(n)$ is a function whose value is never negative then $g(n)T(n)$ is $O(g(n)f(n))$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider Iterated logarithm of n, written $\log^* n$. Defined as below.

$$\lg^* N = \min\{k \mid \underbrace{\lg \lg \cdots \lg}_{k} N \leq 1\}$$

Which of the following is TRUE?

(log base can be considered as any constant)

A. $\lg \lg^* n = o\left(\lg^* \lg n\right)$ here o is small oh
B. $\lg^* \lg n = o\left(\lg \lg^* n\right)$ here o is small oh
C. $\lg^* \lg n = \boldsymbol{\Theta}\left(\lg \lg^* n\right)$
D. $\lg^* \lg n$ and $\lg \lg^* n$ are not comparable

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Arrange the following functions in their increasing order of growth.

(If $\mathrm{f(n) = O(g(n)}$ and $\mathrm{g(n) = O(h((n))}$ then order is $\mathrm{f(n), g(n), h(n)}$.)

In all options $\log^2 n$ means $(\log n)^2$

A. $2^{\log^2 n}, \quad 4^{\log n}, \quad (\sqrt{2})^{\log n}, \quad 5\sqrt{n}$

B. $4^{\log n}, \quad 2^{\log^2 n}, \quad 5\sqrt{n}, \quad (\sqrt{2})^{\log n}$

C. $5\sqrt{n}, \quad (\sqrt{2})^{\log n}, \quad 4^{\log n}, \quad 2^{\log^2 n}$

D. $5\sqrt{n}, \quad 4^{\log n}, \quad 2^{\log^2 n}, \quad (\sqrt{2})^{\log n}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Which of the following prints the last node of a circular (non empty) linked list?

In all options, head is pointing to the first node of the circular (non empty) linked list.

A.

```
struct node *current = head;
while(current->next!=head)
current=current->next;
printf("%d", current->value);
```

B.

```
struct node *current = head->next;
while(current->next!=head)
current=current->next;
printf("%d", current->value);
```

C.

```
struct node *current = head;
while(current!=head->next)
current=current->next;
printf("%d", current->value);
```

D.

```
struct node *current = head;
while(current->data!=head->data)
current=current->next;
printf("%d", current->value);
```

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Consider two programs given below. Consider a non-empty linked list which is either a circular or singly linked list. $\mathrm{P1}$ and $\mathrm{P2}$ are two programs which try to check if it is circular or not. Let head point to the first node of the linked list.

$\mathrm{P1}$ takes the head of the linked list as its argument and $\mathrm{P2}$ takes head as the first argument and head->next as the second argument at the first call.

```
int P1(struct node* head)
{
    if (head==null)
        return 1;

    struct node* next = head->next;

    while(next!=null && next !=head)
    next = next->next;
    return (next == head);

}
```

```
int P2(struct node* head, struct node* cur)
{
    if (head==null)
        return 1;

    if(cur==null)
        return 0;

    if(head==cur)
        return 1;

    return P2(head, cur->next);
}
```

Which of the following is/are true?

A. $\mathrm{P1}$ returns 1 if given linked list is circular
B. $\mathrm{P2}$ returns 1 if given linked list is circular

C. $P1$ returns 1 if given linked list is having even number of nodes
D. $P2$ returns 1 if given linked list is having even number of nodes

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

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